

Data Analytics (DA)

Module -2

Power BI Assignment 1 – Data Transformation & Data Modelling

Dataset Description:

You will work with the following three datasets:

- List of orders

Attributes: Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit • Order Details

Attributes: Order ID, Category, Sub-Category, Quantity, Amount, Profit

- Sales Target

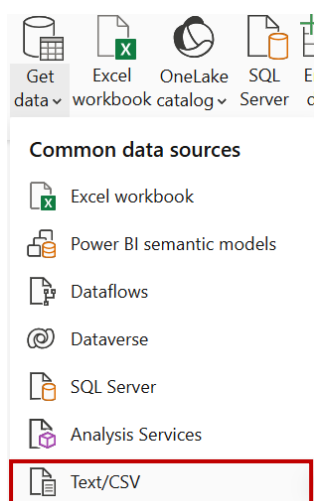
Attributes: Category, Month of Order Date, Target

Tasks

Data Import & Transformation

1. Data Import:

- Import *List of Orders.csv* into Power BI.



List of Orders.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

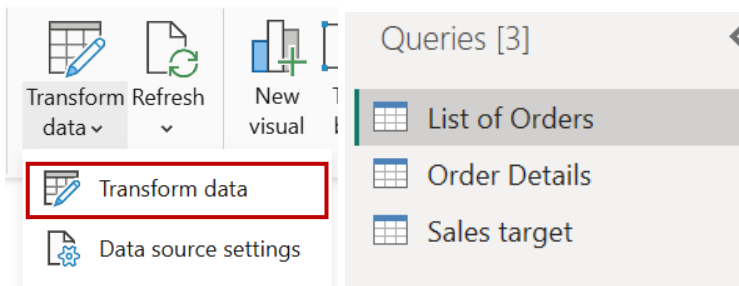
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

- List of Orders
 - City
 - CustomerNa... ..
 - Order Date
 - Order ID
 - State

- Open the list of orders in the Power Query Editor using the Transform Data option.



- Import Order Details.csv and Sales Target.csv into Power Query Editor.

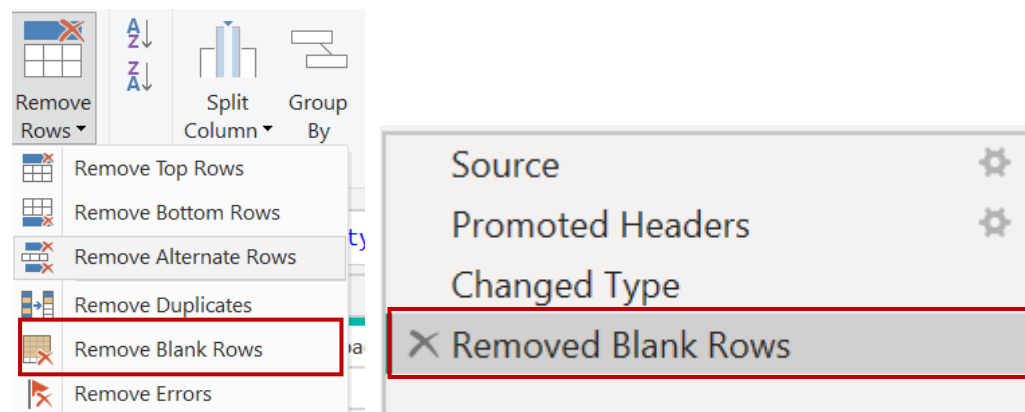
Queries [3] | fx | = Table.TransformColumnTypes("#Promoted Headers",{{"Order ID", type text}, {"Order Date", type date}),

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

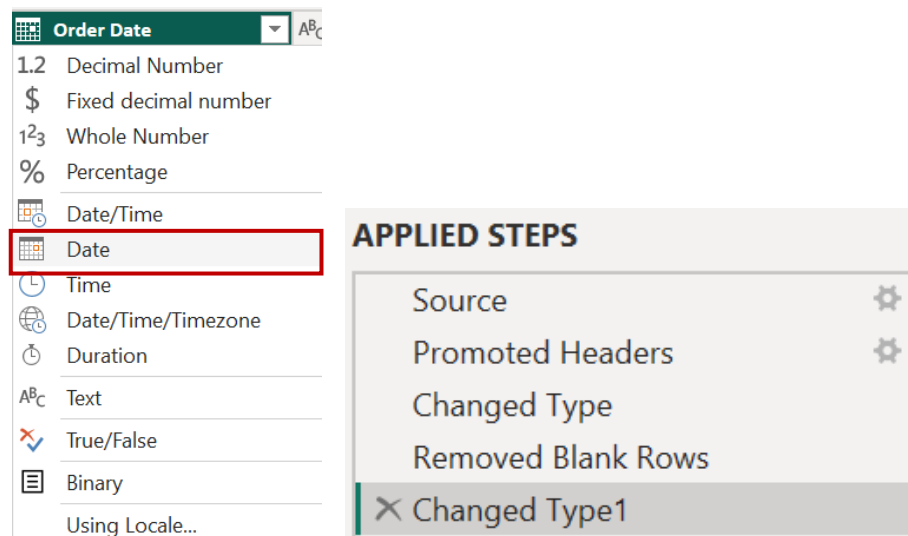
2. Row Limiting & Data Types:

- Restrict the *List of Orders* table to the first 500 rows.

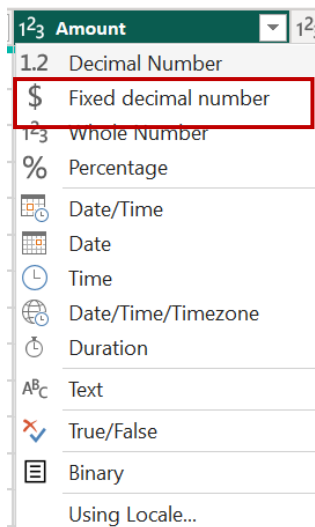
499	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
500	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore
501		null			
502		null			
503		null			



- Set the Order Date column to the Date data type.



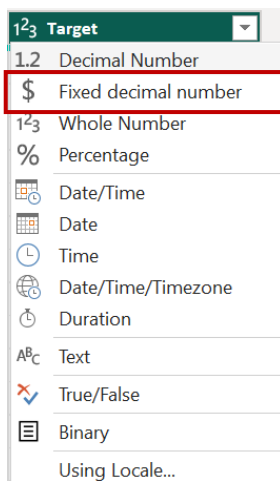
- Change the Amount and Target columns to fixed decimal numbers.



\$ Amount
4.00
6.00
6.00
6.00
7.00

APPLIED STEPS

Source	⚙
Promoted Headers	⚙
Changed Type	
Sorted Rows	
✕ Changed Type1	



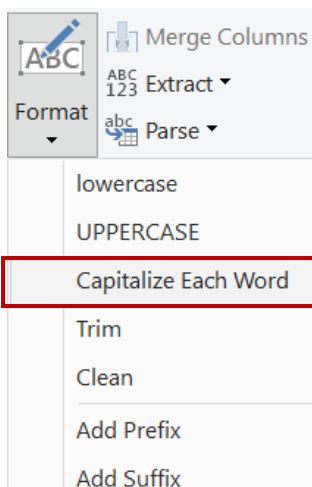
\$ Target
10,400.00
10,500.00
10,600.00
10,800.00
10,900.00
11,000.00
11,100.00
11,300.00

APPLIED STEPS

Source	⚙
Promoted Headers	⚙
✕ Changed Type	

3. Text Formatting:

- Format the **CustomerName** column to **proper case** to ensure consistent capitalization.



APPLIED STEPS

Source	⚙
Promoted Headers	⚙
Changed Type	
Removed Blank Rows	
Changed Type1	
✕ Capitalized Each Word	

4. Column Creation:

- Merge the **City** and **State** columns to create a new column named **Location** in the format: *City, State*

Location = 'List of Orders'[City]&" "&'List of Orders'[State]

Location
Ahmedabad Gujarat
Pune Maharashtra
Bhopal Madhya Pradesh
Jaipur Rajasthan
Kolkata West Bengal
Bangalore Karnataka
Kashmir Jammu and Kashmir
Chennai Tamil Nadu
Lucknow Uttar Pradesh

- Create a custom column named **Profit Margin** calculated as: *Profit / Amount* (expressed as a percentage)

Profit Margin = 'Order Details'[Profit]/'Order Details'[Amount]

Profit Margin
-55.56%
-10.00%
0.00%
-6.25%
12.50%
22.22%
21.05%
31.58%
-90.00%

5. Conditional Column:

○ Add a conditional column named **Profit Status** based on:

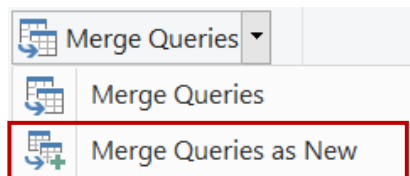
- Profit < 0 → *Loss*
- Profit = 0 → *Break-Even*
- Profit > 0 → *Profit*

```
Profit Status = IF('Order Details'[Profit Margin] < 0, "Loss", IF('Order Details'[Profit Margin] = 0, "Break-Even", "Profit"))
```

Profit Status
Loss
Loss
Break-Even
Loss
Profit
Profit
Profit
Profit
Loss

6. Merging Data (Joins)

○ Merge the *List of Orders* and *Order Details* tables using **Order ID**.



Merge

Select tables and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25697	4.00	-3	1	Clothing	Skirt
B-25656	6.00	3	1	Clothing	Hankerchief
B-26095	6.00	1	1	Clothing	Kurti
B-25655	6.00	-3	1	Clothing	Hankerchief
B-25740	7.00	0	2	Clothing	Skirt

Join Kind
Inner (only matching rows)

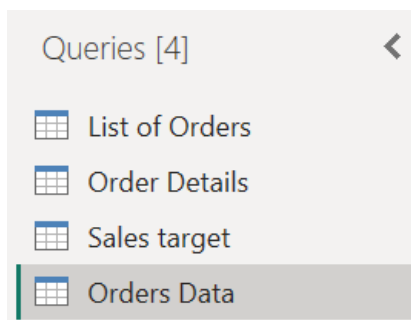
☐ Use fuzzy matching to perform the merge

Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table, and 1500 of 1...

OK Cancel

- Name the merged output table as **Orders Data**.



7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.

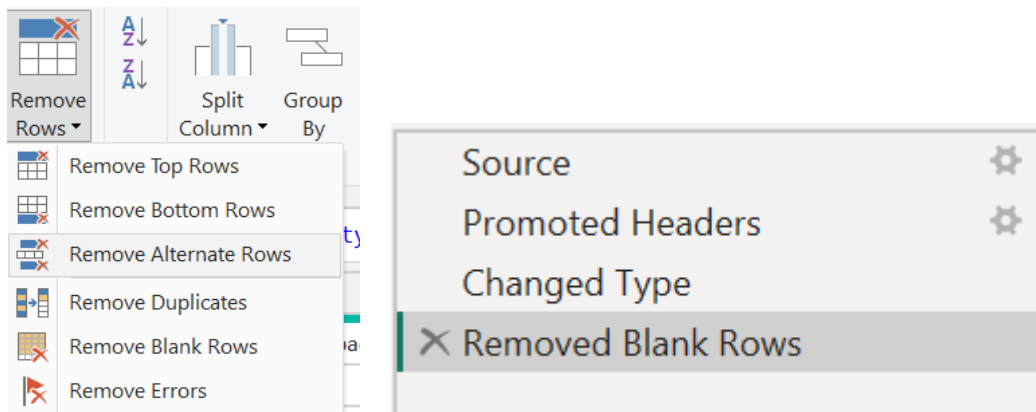
No Missing values are found.

Define and apply appropriate strategies such as:

- Replacing missing values

Mean, Median, Mode can be used for replacing missing values.

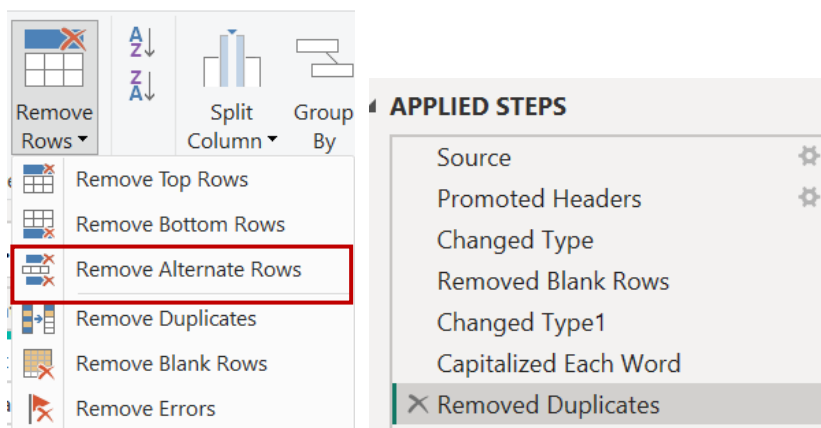
- Removing records



- Retaining nulls with justification

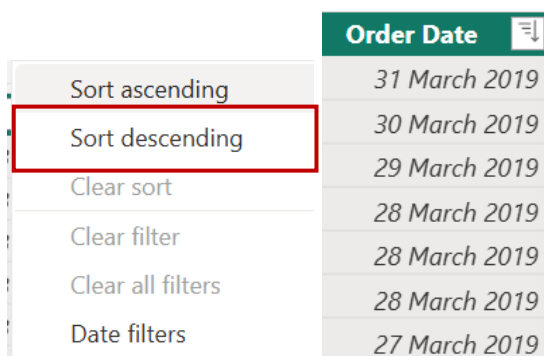
Retaining nulls means keeping missing values in the dataset instead of replacing or removing them when the missing information itself is meaningful or when filling/removing the data could distort the analysis.

- Identify duplicate rows and define a suitable handling strategy based on business logic.



8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.



- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Sort ascending
Sort descending
Clear sort

Clear filter

Clear all filters

Text filters

Search

- ☒ (Select all)
☐ Andhra Pradesh
☐ Bihar
☐ Delhi
☐ Goa
☐ Gujarat
☐ Haryana
☐ Himachal Pradesh
☐ Jammu and Kashmir
☐ Karnataka
☐ Kerala
☐ Madhya Pradesh
☐ Maharashtra
☐ Nagaland
☐ Punjab
☐ Rajasthan
☐ Sikkim
☒ Tamil Nadu
☐ Uttar Pradesh
☐ West Bengal

OK

Cancel

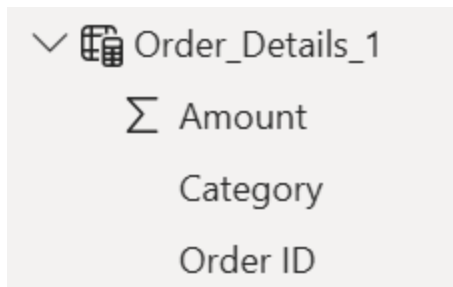
Order ID	Order Date	CustomerName	State	City
B-26081	22 March 2019	Aarushi	Tamil Nadu	Chennai
B-26018	14 February 2019	Aarushi	Tamil Nadu	Chennai
B-26008	09 February 2019	Kalyani	Tamil Nadu	Chennai
B-25860	15 November 2018	Akshay	Tamil Nadu	Chennai
B-25788	21 September 2018	Dinesh	Tamil Nadu	Chennai
B-25716	11 July 2018	Surabhi	Tamil Nadu	Chennai
B-25698	23 June 2018	Amisha	Tamil Nadu	Chennai
B-25608	08 April 2018	Aarushi	Tamil Nadu	Chennai

9. Grouping and Aggregating Data

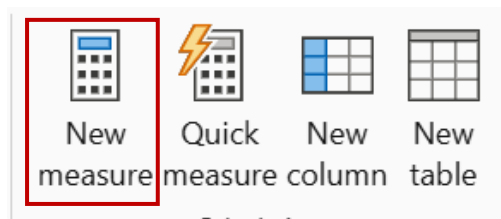
- Duplicate the Order Details table and perform the following:

Order_Details_1 = 'Order Details'

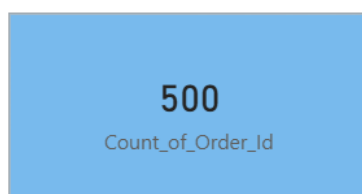
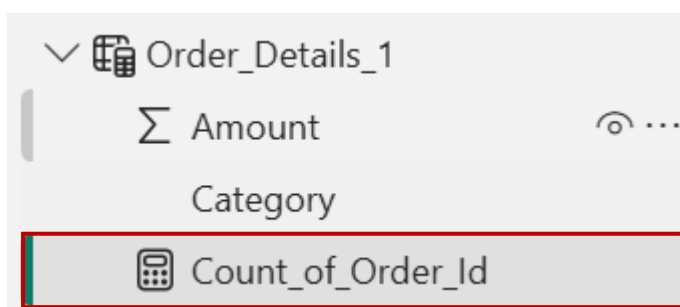
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25656	₹ 6	3	1	Clothing	Hankerchief	50.00%	Profit
B-25873	₹ 7	1	1	Clothing	Hankerchief	14.29%	Profit
B-25851	₹ 10	2	2	Clothing	Hankerchief	20.00%	Profit
B-26088	₹ 11	5	2	Clothing	Hankerchief	45.45%	Profit
B-25979	₹ 12	1	2	Clothing	Hankerchief	8.33%	Profit



o Count of Order ID



Count_of_Order_Id = COUNT(Order_Details_1[Order ID])



o Average Profit by Category

Category	Average of Profit
Clothing	9.91
Electronics	-30.95
Furniture	6.53
Total	4.66

Fields

Category

Average of Profit

o Total Amount by Sub-Category

Sub-Category	Sum of Amount
T-shirt	₹ 2,952
Trousers	₹ 4,617
Tables	₹ 2,603
Stole	₹ 5,197
Skirt	₹ 888
Shirt	₹ 2,277
Saree	₹ 12,082
Printers	₹ 7,824
Phones	₹ 7,706
Leggings	₹ 991
Kurti	₹ 889
Hankerchief	₹ 4,034
Furnishings	₹ 2,583
Electronic Games	₹ 4,446
Chairs	₹ 3,740
Bookcases	₹ 9,952
Accessories	₹ 4,341
Total	₹ 77,122

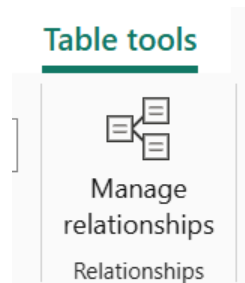
- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Sales_Target_1 = 'Sales target'

Month of Order Date	Sum of Target
Apr-18	₹ 31,400
Aug-18	₹ 33,900
Dec-18	₹ 36,400
Feb-19	₹ 43,600
Jan-19	₹ 43,500
Jul-18	₹ 33,800
Jun-18	₹ 31,600
Mar-19	₹ 43,800
May-18	₹ 31,500
Nov-18	₹ 36,300
Oct-18	₹ 36,100
Sep-18	₹ 34,000
Total	₹ 4,35,900

10. Data Modelling:

- Establish a relationship between:
 - List of Orders and Order Details using Order ID



Select tables and columns that are related.

From table

Orders Data

City	CustomerName	Order Date	Order ID	State
Indore	Hitika	23 April 2018	B-25627	Madhya Prad...
Indore	Ashmi	26 April 2018	B-25637	Madhya Prad...
Indore	Yaanvi	01 May 2018	B-25645	Madhya Prad...

To table

List of Orders

City	CustomerName	Location	Order Date	Order ID	State
Indore	Hitika	Indore Madhy...	23 April 2018	B-25627	Madhya Prad...
Indore	Ashmi	Indore Madhy...	26 April 2018	B-25637	Madhya Prad...
Indore	Yaanvi	Indore Madhy...	01 May 2018	B-25645	Madhya Prad...

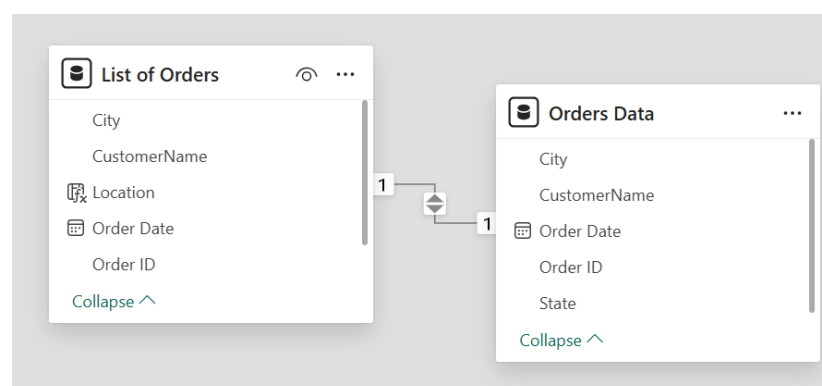
Cardinality

One to one (1:1)

Cross-filter direction

Both

☒ Make this relationship active



- Establish a relationship between:
 - Order Details and Sales Target using Category

← New relationship ×

Select tables and columns that are related.

From table

Order Details ▼

Category	Order ID	Profit	Profit Margin	Profit Status	Quantity	Sub-Categ
Clothing	B-25656	3	50.00%	Profit	1	Hankerc
Clothing	B-25655	-3	-50.00%	Loss	1	Hankerc
Clothing	B-25700	-3	-42.86%	Loss	2	Hankerc

To table

Sales target ▼

Category	Month of Ord...
Furniture	Apr-18
Electronics	Apr-18
Clothing	Apr-18

Cardinality

Many to many (*:*) ▼

Cross-filter direction

Both ▼

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

2. Use Manage Relationships to ensure relationships are active and correctly defined.

Manage relationships ➤

+ New relationship		⚡ Autodetect	✎ Edit	🗑 Delete	≡ Filter ▼
☐ From: table (column) ↑	Relationship	To: table (column)	Status		
☐ Order Details (Category)	* ↔ *	Sales target (Category)	Active	...	
☐ Orders Data (Order ID)	1 ↔ 1	List of Orders (Order ID)	Active	...	